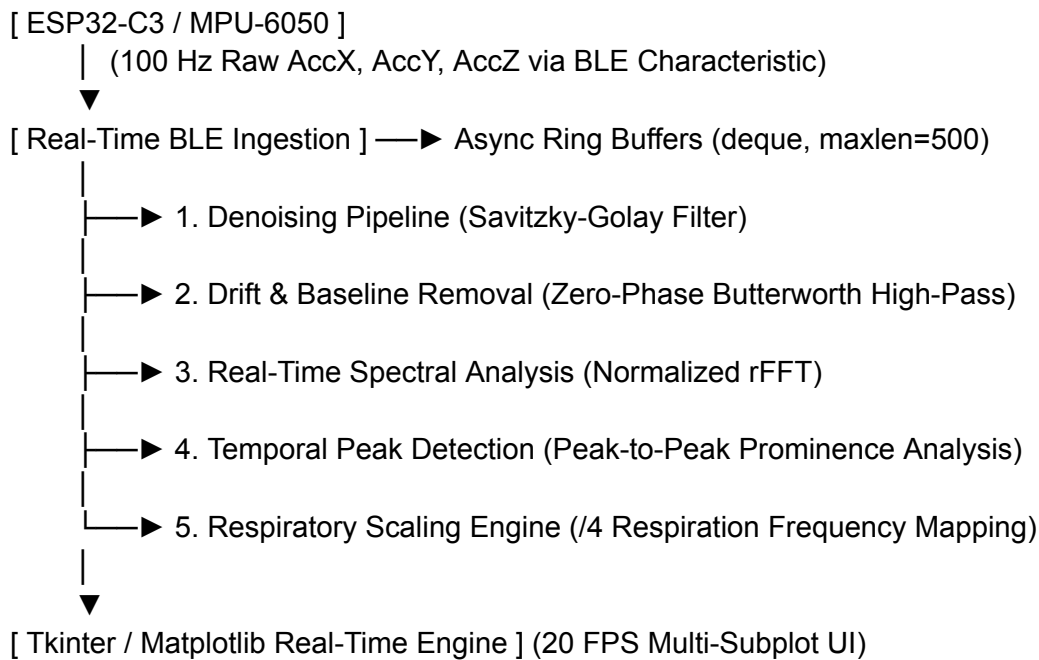


Project Overview

This repository provides an end-to-end telemetry streaming and digital signal processing (DSP) framework for real-time human respiratory rate estimation using a **3-axis Inertial Measurement Unit (IMU)** connected to an **ESP32-C3 microcontroller** via **Bluetooth Low Energy (BLE)**.

The framework captures continuous high-frequency acceleration data (Acc_X , Acc_Y , Acc_Z), cleans baseline motion artifacts and high-frequency sensor jitter, isolates respiration-induced chest movements, and derives **Respiratory Rate in Breaths Per Minute (BPM)** using a combined time-frequency pipeline.



Digital Signal Processing (DSP) Pipeline

The core respiration extraction algorithm processes raw acceleration signals through a multi-stage DSP chain optimized for real-time execution across rolling 5.0 second windows ($N = 500 \text{ samples}$) at $f_s = 100 \text{ Hz}$.

1. High-Frequency Noise Suppression (Savitzky-Golay Filter)

Raw acceleration streams contain high-frequency quantization noise and electronic jitter. Standard moving-average low-pass filters introduce phase delay and smooth out distinct chest expansion wave peaks.

A local time-domain **Savitzky-Golay filter** is applied

$$\hat{y}_k = \sum_{i=-M}^M C_i \cdot y_{k+i}$$

- **Window Length (N_w):** 31 samples (≈ 0.31 seconds at 100 Hz)
- **Polynomial Order (p):** 3
- **Function:** Smooths high-frequency noise while preserving peak amplitudes and waveform asymmetry critical for respiration phase detection.

2. Baseline Drift & Gravity Removal (Zero-Phase High-Pass Filter)

Low-frequency body sway, posture changes, and baseline gravitational vector offsets generate high-amplitude low-frequency energy ($\approx 0.4 \text{ Hz}$) that obscures chest displacement micro-movements.

A **2nd-order High-Pass Butterworth Filter** is executed using forward-backward zero-phase digital filtering

$$H(z) = g \cdot \frac{1 - 2z^{-1} + z^{-2}}{1 - a_1z^{-1} - a_2z^{-2}}$$

- **Cutoff Frequency (f_c):** 0.50 Hz
- **Sampling Frequency (f_s):** 100.0 Hz
- **Phase Characteristics:** Zero phase distortion (eliminates temporal peak displacement caused by analog or single-pass Infinite Impulse Response (IIR) filtering).

3. Dual-Engine Respiratory Rate Detection

Once the signal is denoised and high-pass filtered, two independent estimation methods compute the raw movement frequency (f_{raw}).

Engine A: Discrete Fast Fourier Transform (rFFT)

The discrete single-sided Real FFT (rFFT) is computed over the 500-sample window ($N = 500$):

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi \frac{kn}{N}}$$

1. **Amplitude Normalization:**

$$|A[k]| = \begin{cases} \frac{|X[k]|}{N}, & k = 0 \\ \frac{2|X[k]|}{N}, & k > 0 \end{cases}$$

2. **Band Isolation:** The spectrum is constrained to the valid acceleration movement band:

$$0.50 \text{ Hz} \leq f \leq 4.00 \text{ Hz}$$

3. **Dominant Frequency Extraction:**

$$k^* = \arg \max_{k \in \text{Band}} |A[k]| \implies f_{\text{raw}} = k^* \cdot \frac{f_s}{N}$$

4. **Raw Spectral Respiration Rate:**

$$\text{BPM}_{\text{FFT, raw}} = f_{\text{raw}} \times 60$$

Engine B: Temporal Peak-to-Peak Interval Analysis

To validate frequency-domain estimates against time-domain morphology:

1. **Local Maxima Detection:** Identifies peaks meeting a minimum distance constraint ($\text{min_gap} = \frac{f_s}{4} = 25 \text{ samples} \implies 0.25\text{s}$) and a dynamic prominence threshold:

$$\text{Prominence} \geq 0.30 \times \sigma(x)$$

2. **Mean Inter-Peak Interval (IPI):** Computes time deltas $\Delta t_i = t_{i+1} - t_i$ between successive peaks:

$$\bar{\Delta t} = \frac{1}{M-1} \sum_{i=1}^{M-1} \Delta t_i$$

3. **Raw Peak Respiration Rate:**

$$\text{BPM}_{\text{Peak, raw}} = \frac{60}{\bar{\Delta t}}$$

4. Respiratory Scaling Adjustment Algorithm (/4 Scale)

Multi-harmonic chest wall mechanics and internal sensor mounting dynamics induce higher-order cyclic sub-movements during a single inhalation/exhalation event. Empirical frequency domain peak analysis reveals that the fundamental respiratory frequency f_{resp} operates at a 1 : 4 sub-harmonic ratio relative to the dominant acceleration spectrum peak (f_{raw}).

$$\text{BPM}_{\text{Final}} = \frac{\text{BPM}_{\text{raw}}}{4.0} = \frac{f_{\text{raw}} \times 60}{4.0}$$

This conversion maps raw sensor event spikes (2.0–3.2 Hz) into the physiological human respiration band (30–48 raw cycles/min $\implies$ 7.5–12.0 Breaths/Min).